



Muhammad Adnan Hanif MIE (PAK)

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PROFESSIONAL SUMMARY

Structural engineer with 10+ years of experience in mega-infrastructure projects, sustainable materials research, and teaching. Proficient in research software such as **OpenSees**, **MATLAB**, **SPSS**, and **Design Expert**, while currently expanding into **ML** models. Blends computational analysis with classroom instruction and student mentorship. Adaptable, precise, and ready to take on research challenges in the field of structural engineering.

EDUCATION

Master of Science in Structural Engineering - National University of Sciences & Technology, Pakistan

(Sep 22 – Sep 24), **CGPA: 4.0/4.0 (1st Position)**

- **Thesis:** Assessment of Bond Strength using Recycled Brick Aggregate Concrete for Sustainable Construction.

Bachelor of Civil Engineering - National University of Sciences & Technology, Pakistan

(Sep 15 – Jun 19), **CGPA: 3.58/4.0**

- **FYP:** Enhancement of Cyclic Performance of Bricks using various Confining Techniques.

RESEARCH EXPERIENCE

- **Experimental and Numerical Assessment of Bond Strength of Lap Spliced Steel Bars in Recycled Brick Aggregate Concrete**
 - Designed and tested reinforced concrete beams with recycled brick aggregates under four-point bending
 - Developed nonlinear fiber-based beam models in **OpenSees**; validated with regression analysis (**SPSS**)
- **Performance and RSM-based optimization of eco-friendly concrete incorporating seashell powder and bamboo chips**
 - Performed compressive and flexural strength testing at both ambient and elevated temperatures
 - Statistical optimization using **Design Expert 13** and **SPSS**
- **Supervised four undergraduate Civil Engineering groups on research projects related to sustainable construction materials**

PUBLICATIONS

Journal Articles

- Chatveera, B., Ejaz, A., **Hanif, M. A.** et al (2025). Synergistic effect of recycled E-waste fiber and polyvinyl alcohol on the properties of green concrete incorporating recycled concrete aggregate. *Developments in the Built Environment*, 100722 <https://doi.org/10.1016/j.dibe.2025.100722>
- **Hanif, M. A.** et al. (2025). Experimental and numerical assessment of bond strength of lap-spliced steel bars in recycled brick aggregate concrete. *Arabian Journal for Science and Engineering*, <https://doi.org/10.1007/s13369-025-10684-y>.
- Ejaz, A., **Hanif, M. A.** et al. (2025). Development of nano-calcium carbonate-modified green self-compacting concrete incorporating recycled tempered glass and waste cotton rope fibers. *Journal of Building Engineering*, 114765, <https://doi.org/10.1016/j.jobe.2025.114765>
- Chatveera, B., Ejaz, A., Sua-Iam, G., **Hanif, M. A.** et al. (2025). Workability of Sustainable Cement Sand Mortar Made with Waste Fine Aggregates. *Engineering, Technology & Applied Science Research*, 15(6), 28674-28679, <https://doi.org/10.48084/etasr.13675>.
- Saingam, P., **Hanif, M. A.** et al (2026). Experimental evaluation of sustainable jute-basalt hybrid FRP systems for flexural strengthening of RC beams with variable wrapping and light-weight aggregate replacement. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-47649-6>

Journal Articles – Under Review

- Bloach, M. A., Ejaz, A., Rizwan, M., **Hanif, M. A.** et al. (2025). Experimental & numerical investigation on shear behavior of beams with partial replacement of natural coarse aggregates by recycled brick aggregates. *Journal of Material Cycles & Waste Management*.

Conference Proceedings

- **Hanif, M. A.**, & Ejaz, A. (2025). *Experimental study on strength and workability of sustainable concrete incorporating cotton fibers and marble waste*. 7th Conference on Sustainability in Civil Engineering (CSCE'25 – An International Conference), Capital University of Science and Technology, Islamabad, Pakistan.
- **Hanif, M. A.**, & Sajid, M.A. (2025). *Behavior of concrete with ceramic tile waste and hydrated lime as partial cement replacements*. **3rd International Conference** on climate resilient infrastructure & disaster management, ICCRIDM-2025, University of Engineering and Technology (UET) Peshawar, Pakistan

TECHNICAL PROFICIENCIES

- **Research Software:** OpenSees, MATLAB, SPSS, Design Expert
- **Structural Analysis & Design Software:** ETABS, ForteWeb, AutoCAD

PROFESSIONAL EXPERIENCE

Lecturer – National University of Sciences & Technology (NUST), Risalpur, Pakistan – Aug 24 - Till Date

- **Teaching Undergraduate courses:** Mechanics of Solids, Structural Analysis, Reinforced Concrete Design
- Supervising **final year design projects** and actively engaged in **curriculum development**.

Deputy Project Manager - Frontier Works Organization, Pakistan – Nov 20 – Sep 22

- Oversaw mega RCC dam project, hydroelectric works, and bridge construction
- Directed large-scale site operations (500+ machinery units, 21 km site) and coordination with international consultants

ACHIEVEMENTS

- **Best Paper Award** at 3rd International Conference on climate resilient infrastructure & disaster management, ICCRIDM-2025, Pakistan
- Recipient of Merit Certificate for securing **First Position** in Master of Science in Structural Engineering (2024)

REFERENCES

- **Associate Professor & HoD Dr. Muhammad Shahid Siddique** – Current HoD - mshahidsiddique@mce.nust.edu.pk
- **Assistant Professor Dr. Ali Ejaz** – MS Research Supervisor - ali.ejaz@nit.nust.edu.pk
- **Assistant Professor Dr. Mamoon Shaafaey** – Technical Committee Member for MS Research - mamoon@nit.nust.edu.pk

All references and certificates will be made available upon request.